

Revisitation analysis

2025-11-30

Table of contents

recurse package	1
Getting started	2
Load packages	2
Load data	3
Recurse analysis	3
Plot revisitation stats	4

recurse package

The **recurse** R package can be used to analyse revisitation patterns in animal movement data. It provides functions to identify and quantify the number of times an animal returns to specific locations, as well as the duration of stays at those locations. This can be useful for understanding habitat use, foraging behavior, and territoriality.

Recurse vignette (tutorial)

<https://cran.r-project.org/web/packages/recurse/vignettes/recurse.html>

Key reference

Bracis, C., Bildstein, K. L., & Mueller, T. (2018). **Revisitation analysis uncovers spatio-temporal patterns in animal movement data**. *Ecography*, 41(11), 1801–1811. <https://doi.org/10.1111/ecog.03618>

Application paper

Campbell, H. A., Loewensteiner, D. A., Murphy, B. P., Pittard, S., & McMahon, C. R. (2020). **Seasonal movements and site utilisation by Asian water buffalo (*Bubalus bubalis*) in tropical savannas and floodplains of northern Australia.** *Wildlife Research* , 48(3), 230–239. <https://doi.org/10.1071/WR20070>

Getting started

Load packages

```
# install.packages("recurse")
```

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
```

```
v dplyr      1.1.4      v readr      2.1.5
```

```
v forcats    1.0.0      v stringr    1.5.2
```

```
v ggplot2     4.0.0      v tibble     3.3.0
```

```
v lubridate   1.9.4      v tidyr      1.3.1
```

```
v purrr       1.1.0
```

```
-- Conflicts ----- tidyverse_conflicts() --
```

```
x dplyr::filter() masks stats::filter()
```

```
x dplyr::lag()     masks stats::lag()
```

```
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to beco
```

```
library(sf)
```

```
Linking to GEOS 3.13.0, GDAL 3.8.5, PROJ 9.5.1; sf_use_s2() is TRUE
```

```
library(move)
```

```
Loading required package: geosphere
```

```
Loading required package: sp
```

```
Loading required package: raster
```

```
Attaching package: 'raster'
```

```
The following object is masked from 'package:dplyr':
```

```
select
```

```
library(recurse)
```

Load data

```
### Load raw data
dingo_data <- read_csv("../data/tanami_collars.csv") %>% filter(x > 0)
```

Rows: 61359 Columns: 53

-- Column specification -----

Delimiter: ","

chr (13): dogname, sex, mine_away, gmt_date, gmt_time, cst_date, cst_month...

dbl (38): x, y, dog_id, attempt_id, fix_id, cstmonth, cstmonth_cnt, csthour...

time (2): cst_time, timemaxwindgist

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
dingo_data_sf <- st_as_sf(dingo_data, coords = c("x", "y"), crs = 4326)

# reproject the data into a projected CRS (UTM zone 52)
dingo_data_sf <- st_transform(dingo_data_sf, crs = 28352)

# convert back to dataframe
dingo_data <- dingo_data_sf %>%
  mutate(x = st_coordinates(geometry)[,1],
         y = st_coordinates(geometry)[,2]) %>%
  st_set_geometry(NULL)

# create a timestamp column
dingo_data$timestamp <- as.POSIXct(paste(dingo_data$gmt_date, dingo_data$gmt_time),
                                   format = "%d/%m/%Y %H:%M:%S",
                                   tz = "UTC")

# convert to local time
dingo_data$timestamp <- with_tz(dingo_data$timestamp, tzone = "Australia/Darwin")
```

Recurse analysis

```
# select columns for recurse
dingo_recurse <- dingo_data %>%
  mutate(id = dogname,
         t = timestamp) %>%
  dplyr::select("x", "y", "timestamp", "id")

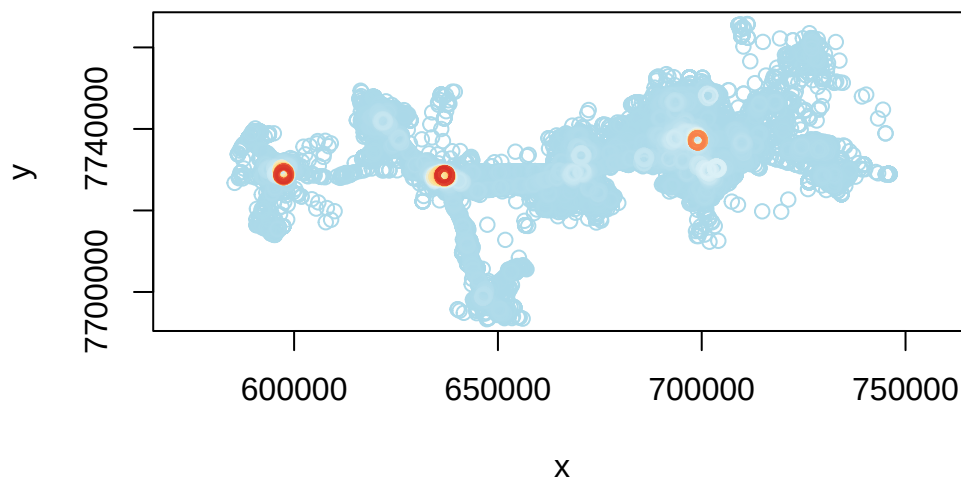
# convert to dataframe
dingo_recurse <- as.data.frame(dingo_recurse)

# calculate revisitations
dingo_recurse_object <- getRecursions(dingo_recurse, radius = 500,
                                     timeunits = "hours")

# extract the revisitation stats
revisit_stats <- dingo_recurse_object$revisitStats
```

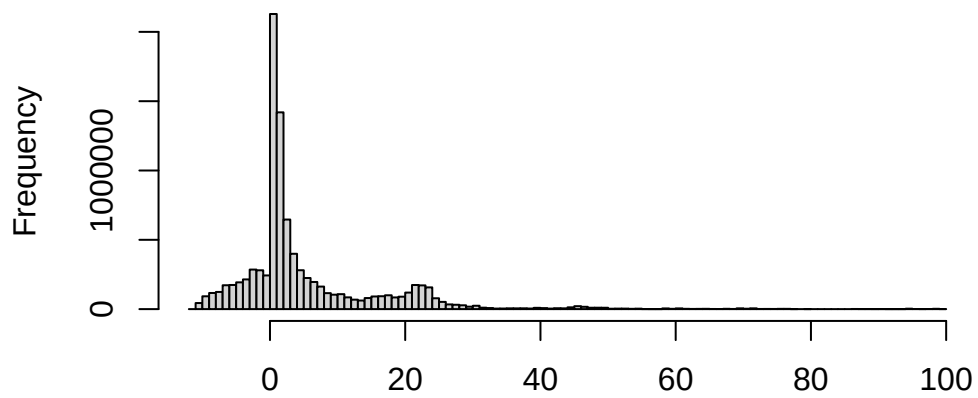
Plot revisitation stats

```
# plot the revisitations
plot(dingo_recurse_object, dingo_recurse)
```



```
hist(revisit_stats$timeInside[revisit_stats$timeInside < 100],
     breaks = 100,
     main = "Time inside (hours)")
```

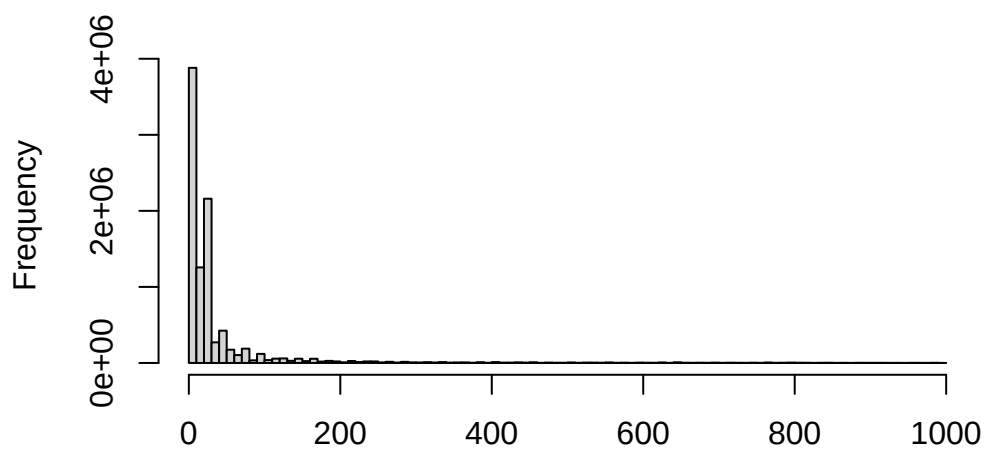
Time inside (hours)



```
revisit_stats$timeInside[revisit_stats$timeInside < 100]
```

```
hist(revisit_stats$timeSinceLastVisit[revisit_stats$timeSinceLastVisit < 1000],  
     breaks = 100,  
     main = "Time since last visit (hours)")
```

Time since last visit (hours)



```
revisit_stats$timeSinceLastVisit[revisit_stats$timeSinceLastVisit < 1000]
```